

Continue " Project Status Snapshot

Revision: 19 **Last modified:** 2026-06-16T23:55:00Z **Session:** 2026-06-16 (Session 14 " search-broken/crashing fixed: multi-word encoding storm, private-tracker cookie auth, deploy pipeline, healthz, kickass; PROVEN on nezha " 2600 results, all 4 private trackers auth, none crashed. In-flight: P0 UTF-8/Cyrillic encoding fix) **Last commit:** 646b295 (Status ledger Rev 7 " proven search/auth/healthz fixes Â§11.4.153) **Branch:** main **Working tree:** has the in-flight UTF-8 RED test untracked (tests/unit/test_plugin_unicode_query_encoding.py); search/auth/deploy fixes all published at HEAD

Send continue to pick up exactly where we left off. This file is the single source of truth for session handoff (Â§12.10 / Â§11.4.131).

CURRENT STATE " Session 14 (2026-06-16): search fixed end-to-end + proven on nezha; one P0 in flight

Operator-reported "search is broken / crashing a lot" + "rutracker/nnmclub never authenticate" " ROOT-CAUSED and FIXED, proven live on the distributed-boot production stack (nezha.local, podman). Definitive end-to-end proof: full-fleet the matrix on nezha returned **2600 results, 23/29 trackers contributing, encoding-crashed: NONE, all four private trackers authenticated** (rutracker 50 / nnmclub 50 / kinozal 50 / iptorrents 49). QA evidence: docs/qa/search-fix-verify-20260616/ (results.md + html/pdf/docx). BUGFIXES.md Rev 19 entries 48"55.

What landed (Session 14 " all published at HEAD 646b295)

- **Multi-word query URL-encoding crash, 17 nova3 plugins** (dbd3858 7 maintained + da7d709 10 adopted) " a literal space crashed urllib (URL can't contain control characters); single-word worked, multi-word silently dropped ~17 plugins " "search broken / crashing". Fix: per-plugin percent-encoding. **Verified live** (nezha): scoped the matrix 592, full-fleet 2600, NONE crashed. (BUGFIXES #48)
- **plugin_bad_query_encoding honesty classifier** (33d90f2) " stopped reporting a self-inflicted encoding bug as a remote plugin_crashed (Â§11.4.6). (BUGFIXES #49)
- **Private-tracker cookie auth** " RUTRACKER_COOKIES injection bypasses the CAPTCHA-walled login (2fc29fc) + existing NNMCLUB_COOKIES; cookies via scripts/extract-tracker-cookies.sh (Â§11.4.10). **Verified 50/50** on nezha. (BUGFIXES #50)
- **/auth/status cookie reflection** (9c2f8dc) " chips now show cookie-authenticated trackers green before the first search. (BUGFIXES #51)
- **JSON /api/v1/healthz** (137d7ff) " was swallowed by the SPA catch-all (returned HTML); now JSON. (BUGFIXES #52)
- **scripts/deploy-remote.sh** " **4 silent-failure bugs** (d5b58cb / 9e059d3 / 42cdb02 / e6b9f8f) that kept fixes from landing on the remote (Â§11.4.108): inline-YAML-comment path, rsync abort on container-owned config/, py_compile false "Syntax: Invalid", and install-plugin targeting only qbittorrent not qbittorrent-proxy. Pipeline now completes [1/5] " [5/5] clean; re-verified via the pipeline (not manual cp). (BUGFIXES #53)

- **kickass** §11.4.112 **Won't-fix** (f3e7a4f) 403 behind a Cloudflare/JS challenge, structurally unsolvable by a headless nova3 plugin; documented in docs/research/. (BUGFIXES #54)
- **Regression baseline:** full unit suite at HEAD **4277 passed, 0 failed** (up from 4216 pre-session, zero regression).

Host note

- **Host disk-full** observed this session (T7 // Volumes/T7/tmp is the temp dir for this work). Keep an eye on free space before any container rebuild or large deploy a full disk silently fails rsync/builds.

In flight "NEXT ITEM (P0, do this first)

- **UTF-8 / Cyrillic query crash in 15 plugins (OPEN "NOT yet fixed, §11.4.6).** A sibling defect class to #48 surfaced during the #48 discovery extend-pass (§11.4.146/§11.4.118): the percent-encoding fix addressed the literal-space control-character crash; the non-ASCII (Cyrillic) byte path is a DISTINCT crash. RED reproduction test is in the working tree (untracked): tests/unit/test_plugin_unicode_query_encoding.py. **NEXT:** complete per-plugin UTF-8-safe encoding across the 15 affected engines, flip RED→GREEN (§11.4.115), run the §11.4.146 extend-pass to enumerate the full affected set, verify live on nezha, then complete BUGFIXES #55 with the GREEN proof + commit hash.

CURRENT STATE "Session 13 (2026-06-13): stack booted + qBittorrent 5.x compat

The #1 release blocker is CLEARED. The full multi-container stack is BOOTED and the live detect→send→torrent round-trip is GREEN "the synthetic infohash is independently confirmed in qBittorrent's real torrent list, then cleaned up (§11.4.14).

- **Boot blocker (macOS/podman) "root-caused + fixed.** /Volumes/T7 (external SSD) was not shared into the podman applehv VM → podman create hung on an unreachable bind path (proven: py3.13 + 3.14 hung identically; NOT a python-version issue). Fix (operator-authorized §11.4.122): recreated the machine with --volume /Volumes/T7:/Volumes/T7 --cpus 4 --memory 6144; export QBITTORRENT_DATA_DIR=/Volumes/T7/Projects/Boba/downloads; ./start.sh. All 5 containers healthy. Earlier boba-ctl up -d boot bug fixed in 8b6d245.
- **3 REAL product defects (qBittorrent v5.2.1) "0dc247b.** The proxy spoke legacy text (200 "Ok. ") while linuxserver:latest speaks 5.x JSON/204. download-proxy/src/api/routes.py: _qbit_login_succeeded (204+QBT_SID cookie, all 4 login sites) + _qbit_add_succeeded (modern JSON add + 409-duplicate idempotency, all 3 add sites). Guard: tests/unit/test_qbit_login_compat.py (19 tests). Independent code-review (§11.4.142/§11.4.134): 2 BLOCKING + 1 warning caught → clean GO. Documented BUGFIXES.md Rev 14 entries 36→38.
- **macOS LAN access "ef121e7:** scripts/ensure-macos-tunnel.sh now supports TUNNEL_BIND_ADDR=0.0.0.0 for LAN-IP access (default 127.0.0.1 unchanged).
- **Manual testing (stack UP):** http://localhost:7187/ (dashboard) → http://localhost:7186 (qBittorrent admin/admin) → :7189 boba-jackett → :9117 Jackett. Re-tunnel after reboot:

bash scripts/ensure-macos-tunnel.sh (TUNNEL_BIND_ADDR=0.0.0.0 for LAN).

- **Remaining (operator-gated):** headful MV3-load e2e (needs a real display); store assets/submission.
 - **Process note:** a background integration suite (test_live_containers.py, which does compose down/up) tore down the live stack mid-session; killed + restored. Parallel work during operator testing MUST stay hermetic (Â§11.4.119) â€” no integration/e2e/container-touching tests.
-

CURRENT STATE â€” Session 10 (BobaLink browser extension)

New major feature in flight: the BobaLink browser extension. Plan + discovery + Phase 1 foundation + backend BE-1/BE-2 + a token-auth security fix landed. The Python unit suite stays FULLY GREEN at 4149 passed.

What is BobaLink

A WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs on any page and forwards them to the running Boba merge service on port **7187**. Backend contract: POST http://<host>:7187/api/v1/download. The full plan, discovery analysis, and traceability live under docs/browser_extension/ â€” master plan IMPLEMENTATION_PLAN.md (9 phases), analysis artifacts _analysis/01â€”06, planning artifacts _plan/Aâ€”F, and the 245-item traceability matrix _plan/C-traceability-matrix.md (240 v1 / 5 v2, proving nothing is skipped per Â§11.4.118).

What landed (Session 10 â€” BobaLink)

- b2356ae â€” BobaLink master implementation plan + discovery/planning artifacts (docs/browser_extension/); also bumped HelixQA submodule bcac236 â†’ 4d2dcb2.
- 33a9815 â€” extension **Phase 1 foundation** scaffolded (extension/ â€” WXT config, TS, shared libs/types under extension/src/); **71 anti-bluff Vitest unit tests** (vitest 71 passed, eslint clean â€” crypto imports the REAL module, not an inline copy).
- d46bffb â€” backend **BE-1** (CORS for extension origins) + **BE-2** (raw .torrent upload endpoint) added to the Python :7187 API.
- 284d1c4 â€” bumped HelixQA submodule 4d2dcb2 â†’ bca3b36; new Challenge bank submodules/helixqa/banks/boba-bobalink.yaml (6 cases).
- 192b945 â€” **security fix:** env-gated shared-secret BOBA_API_TOKEN enforced on the three :7187 download-write endpoints (when the env var is set, requests must carry a matching X-Boba-Token / Bearer token; behaviour unchanged when unset). Verified by **15 token-auth tests** (tests/unit/api_layer/test_download_token_auth.py).

What landed since Session 10 (BobaLink â€” Phases 2 & 3 + Phase 4 leaf)

- 7225470 â€” Phase 2 wave-1: parsers (bencode.ts/magnet.ts) + scanner base/site-db.
- fa03323 â€” Phase 2 wave-2: .torrent-file SHA-1 infohash + link/text scanners + perf/stress.
- 946c61e â€” Phase 2 complete: scanner orchestrator (cross-scanner dedup).

- 2e59572 â€” lint test files clean; lint script extended to tests/.
- e8fde43 â€” Phase 3 shell + Phase 4 api leaf: api / {boba-client, queue} + content/popup/options.
- 15a9a61 â€” Phase 3 capstone: background service worker (message router). **HEAD.**

Suite status

- Full Python unit suite: **4149 passed** â€” `pytest tests/unit/ --import-mode=importlib` (Session 10 baseline; unchanged this session). Evidence: qa-results/tokenauth_fullsuite_*.log.
- Extension Vitest corpus: **379 passed across 33 spec files** â€” same-session `npx vitest run` (Tests 379 passed (379)); `tsc --noEmit clean`; `npm run lint 0/0.` (+92 over the 287/22 baseline.) Build: `npx wxt build â†’ loadable .output/chrome-mv3/` (Â§11.4.38 â€” 8/8 manifest assets verified present).

Current phase + next steps (Session 11 â€” two parallel waves landed; reviewed checkpoint)

- **Phases 1, 2, 3 + extension shell + background SW: COMPLETE.**
 - **WXT build wiring: COMPLETE** â€” endpoints at `src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html}` (thin wrappers); `npx wxt build â†’ loadable .output/chrome-mv3/`; matches derived from `SITE_SELECTORS` (24 hosts, no `<all_urls>`); least-privilege manifest; Â§11.4.38 verified.
 - **Phase 4 (backend integration): IN PROGRESS** â€” api leaf @e8fde43; **Phase-7 decrypt-before-send WIRED** â€” `BobaClient.create()` decrypts the encrypted `BobaApiToken` bundle; background reads the session passphrase from `chrome.storage.session`, sends decrypted plaintext, default-open when locked (REDâ†’GREEN; token/passphrase never logged). **PENDING:** live-7187 integration (`require_backend(7187)`) + detectâ†’sendâ†’torrent-in-qBittorrent E2E on the real backend.
 - **Phase 5 (tab groups): IN PROGRESS** â€” standalone `src/tabgroups/index.ts` (dedupe-across-group + batch dispatch) + 13 tests landed; **PENDING:** wire into background `MENU_SEND_GROUP` + add `tabGroups/tabs` perms (least-privilege review first).
 - **Phase 6 (i18n): IN PROGRESS** â€” `locale.test.ts` guards en-catalog completeness. **Phase 7 (security): IN PROGRESS** â€” decrypt path + `tests/security/*` (least-privilege/CSP/no-hardcoded-secret/secret-storage). **Phase 8: IN PROGRESS** â€” 379/33 green; real-module Challenge challenges/extension/detect_and_forward_challenge.sh (mutation-verified); HelixQA BOBA-LINK-007; e2e is an honest operator-gated SKIP (sandbox canâ€™t load extensions). **Phase 9: PENDING** â€” `ci-ext.sh` gate + per-store packaging + Â§11.4.65 doc siblings.
 - **Next actions:** (1) **independent-subagent code-review re-pass** for the correctness/security/build lenses once the platform subagent-dispatch throttle clears (done in-context this session; re-run before any release tag per Â§11.4.40); (2) Phase 5 integration into background + perms; (3) Phase 4 live-7187 integration + E2E; (4) Phase 9 packaging; (5) regenerate `Status_Summary.md` + Â§11.4.65 HTML/PDF siblings for the browser_extension Status docs. Status: docs/browser_extension/Status.md (Rev 2, accurate). QA evidence: docs/qa/bobalink-2026-06-10-session11/.
-

CURRENT STATE â€” Session 10 (2026-06-10, subagent-driven loop)

Suite stays FULLY GREEN. This session was hardening + verification, subagent-driven (Â§11.4.70), every change code-reviewed before commit (Â§11.4.125/Â§11.4.134).

What landed (Session 10)

- `test_credential_env_wiring.py` **isolation fix** â€” resolved the Session-9 â€œknown low-priority follow-upâ€” below. A `merge_service` fake-namespace stub was installed at module-collection time with no teardown, so `conftest`â€™s `_isolate_download_proxy_modules` captured-and-restored the *fake* stub into siblings in some run orders. Fixed by moving stub install into a `search_mod` fixture that snapshots/restores the 3 injected keys (`merge_service`, `.search`, `.retry`) in finally. **Evidence:** RED reproduction (the 5 credential tests + a temporary leak-probe) went 1 failed â†’ all green; the fileâ€™s own **5 tests pass** standalone; full unit suite **4121 passed** — seeds default/42/31337 (Session-10 run). Code-review **GO** (proved negation; no weakened assertions; ruff clean).
- `docs/qa/BOB-008/operator_runbook.md` â€” copy-pasteable operator unblock procedure for BOB-008 (cookie path preferred; CAPTCHA path documented with its `cap_sid/cap_code_field` friction), every endpoint claim cited to `auth.py:line` (Â§11.4.83/Â§11.4.99).
- **Latent-leak discovery sweep (Â§11.4.118)** â€” audited all of `tests/` for the module-level fake-stub-no-teardown pattern. Finding: the pattern is a **structural no-op wherever the test dir has an `__init__.py`** (real package pre-loads â†’ `setdefault` never installs the fake), so the ~26 `tests/unit/merge_service/` matches and `test_tracker_stats.py` are **inert** (no RED reproduces; strict TDD â†’ no fix). The single genuinely-uncovered root is `pirateiro` (see follow-up).
- **Constitution inheritance verified (Â§11.4.32/Â§11.4.35)** â€” PASS: pointer present in both layers, propagated anchors present in canonical source, no conflict markers, `pre_build_verification.sh` 18/18 green.
- **pirateiro test-isolation fix** â€” **BOB-063 (operator-approved, implemented)** â€” `tests/unit/test_plugin_pirateiro.py` injected `sys.modules['pirateiro']` at module scope with no teardown; `pirateiro` was the one root uncovered by `conftest`â€™s isolation, so it leaked into later tests. The naive â€œadd to `_POLLUTING_ROOTS`â€” does NOT work (the leak is already inside each per-test snapshot); the real fix caches+re-registers+purges the stub per unit test. **Evidence (Â§11.4.115):** RED 1 failed, 44 passed â†’ GREEN 45 passed; negation proof (disable the purge â†’ re-fails); full suite **4122 passed** — seeds default/42. Standing regression guard `tests/unit/test_pirateiro_isolation_guard.py` (Â§11.4.135). QA at `docs/qa/BOB-063/evidence.md`. **Closed BOB-063** (Task â†’ Completed).
- **Constitution submodule advanced 60e2d66 â†’ f26368b (Â§11.4.26, clean fast-forward, Â§11.4.113-safe)** â€” pulled Â§11.4.140 v2 (BACKGROUND action), Â§11.4.142 (universal code-review â€” already actively enforced this session; added to CLAUDE.md propagated clauses), Â§11.4.143 (video-streaming real-user-journey â€” latent/N-A to this torrent-proxy project). Cascade gate **18/18 green** at the new pin.

Open queue (Issues.md)

- **BOB-008** “RuTracker CAPTCHA” **OPERATOR-BLOCKED**. Unblock procedure documented at docs/qa/BOB-008/operator_runbook.md (preferred: paste bb_session via POST /api/v1/auth/rutacker/cookie-login).
- Everything else closed (DB “i,ŽMD in sync, **63 items**, bin/workable-items validate OK).

Known low-priority follow-up

- ~~pirateiro-test-isolation-defense-in-depth~~ “**RESOLVED this session** (BOB-063, see above).
- ~~test_credential_env_wiring/env_loader_stub-teardown-quirk~~ “**RESOLVED this session** (see above).

CURRENT STATE “Session 9 (2026-06-10, overnight autonomous loop)”

The unit suite is now **FULLY GREEN** and **DETERMINISTIC**.

Metric	Value
Unit tests	4121 passed, 0 failed “pytest tests/unit/ --import-mode=importlib (5m24s)
Determinism (Â§11.4.50)	top-level scope 3150 passed identical across --randomly-seed 7/42/100/31337/12345
Commits this session	6e15a8d (crash guards + async/loop hangs), 6230865 (test-pollution + network timeouts) “both pushed to all upstreams
Code review	GO (zero findings/warnings) after Â§11.4.134 iterate-until-GO

What landed (Session 9)

- **8 product fixes** (6e15a8d): degenerate-input crash guards for tokyotoshokan/kickass/yts/piratebay; enricher full-suite-hang fix (aiohttp.ClientTimeout “6); kickass/bitsearch/torrentgalaxy unbounded-loop caps (MAX_PAGES=50); mutation-scanner .venv scope fix; +7 download_proxy tests. Closed **BOB-060**.
- **Test-suite stabilization** (6230865): eliminated all Â§11.4.50 order-dependent pollution “tests/conftest.py _CORRECT_MS_PATH repo-root fix (the dominant bug: corrupted merge_service.__path__ “broke 11 tests incl.Â all scheduler_api), extended _POLLUTING_ROOTS, + per-test socket.socket & os.environ snapshot/restore. Network-I/O timeout hardening (search.py/routes.py/helpers.py/eztv.py). Closed **BOB-061**, **BOB-062**.

Open queue (Issues.md)

- **BOB-008** "RuTracker CAPTCHA" **OPERATOR-BLOCKED**: needs you to complete the CAPTCHA at `/api/v1/auth/rutacker/captcha+ /login`, OR paste a fresh `bb_session` cookie via `/auth/rutacker/cookie-login`. Cannot be solved autonomously.
- Everything else closed (DB "i,ŽMD in sync, 62 items, bin/workable-items validate OK).

How to run the suite (both work now)

- Monolithic: `.venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib` 4121 passed.
- Per-scope (faster, parallelizable): `tests/unit/merge_service/ (801) tests/unit/api_layer/ (170) tests/unit/*.py` top-level (3150).
- Always pass `--import-mode=importlib` (the `merge_service.deduplicator` lazy import needs it; a bare per-file run errors).

Known low-priority follow-up (NOT blocking " suite is green)

- A residual `test_credential_env_wiring / env_loader sys.modules-stub` ordering quirk surfaces only in narrow isolated file-combos; it does NOT manifest in the full top-level scope (5 seeds green) because the new `env/sys.modules` isolation covers it there. Tighten the `env_loader` stub teardown if it ever resurfaces.

Session 8 Summary

Metric	Value
Test files	47 test modules (222 <code>test_*.py</code> files total across all suites)
Test cases	4,074+ passed
Coverage (unit)	88%+ (gate: 49%)
Bugs fixed	21 (BOB-001 through BOB-025, minus BOB-008)
Submodules aligned	constitution, challenges, containers, helixqa, jactett
Ruff / Mypy	pre-existing warnings only, 0 new

Bug Tracker

21 bugs fixed across Session 8: - 10 B-substring parsing fixes - 3 import re missing fixes - 2 comma-size parsing fixes - 2 crash guards - 1 bt4g hang fix - 1 dedup circular import fix - 1 `env_loader` flaky test fix - 1 pirateiro full test suite (44/44)

1 remaining: BOB-008 (RuTracker CAPTCHA) "operator-blocked, needs manual cookie paste.

Test File Inventory

tests/unit/merge_service/ (31 files)

- test_cors_config.py
- test_dead_tracker_bucket.py
- test_deadline_tunable.py
- test_deduplicator.py
- test_deduplicator_edge.py
- test_diag_no_stale_leakage.py
- test_edge_case_challenges.py
- test_enricher.py
- test_enricher_edge.py
- test_hooks.py
- test_html_parsers.py
- test_jackett_autoconfig.py
- test_nnmclub_session_login.py
- test_private_tracker_html_fixtures.py
- test_public_plugin_harness.py
- test_public_plugin_harness_broad.py
- test_public_tracker_capture.py
- test_public_tracker_subprocess_timeout.py
- test_quality_detection.py
- test_scheduler.py
- test_scheduler_coverage.py
- test_search_concurrency.py
- test_search_coverage.py
- test_search_deep_coverage.py
- test_search_error_paths.py
- test_session_encryption.py
- test_theme_endpoint.py
- test_tracker_stats.py
- test_ttl_caches.py
- test_validator.py
- test_validator_coverage.py

tests/unit/api_layer/ (14 files)

- test_auth_coverage_extra.py
- test_concurrent_writers.py
- test_cors_config.py
- test_hooks_coverage.py
- test_hooks_endpoints.py
- test_hooks_remaining.py
- test_nnmclub_auth_endpoints.py
- test_routes_coverage.py
- test_sse_disconnect.py
- test_sse_token_auth.py
- test_theme_state_coverage.py
- test_theme_stream.py
- test_tracker_stats_sse.py

tests/unit/ (55 files, infrastructure + plugin test
test_auth.py, test_auth_coverage.py, test_auth_models.py
test_config.py, test_main.py, test_dashboard.py)

test_content_type_refinement.py
test_credential_env_wiring.py
test_download_proxy_coverage.py, test_download_proxy_deep.py
test_download_merged.py
test_env_loader.py
test_freeleech.py
test_graceful_shutdown.py
test_helpers.py
test_log_filter.py
test_merge_trackers.py
test_novaprinter.py
test_private_tracker_search.py
test_public_tracker_subprocess.py
test_retry_policy.py
test_routes.py, test_routes_coverage.py
test_scheduler_api.py
test_sorting_weights.py
test_sse_disconnect.py
test_streaming.py
test_theme_injector.py, test_theme_wiring.py
test_webui_theme_injector.py
test_ui_module.py, test_ui_sorting.py
test_plugin_*.py

(39 plugin modules: eztv, piratebay, s
limetorrents, torlock, gamestorrents,
anilibra, torrentgalaxy, yts, rutor,
snowfl, torrentdownload, linuxtracker
nmclub, rutracker, megapeer, jackett
one337x, extratorrent, torrentfunk, t
therarbg, academictorrents, ali213, y
glotorrents, pctorrent, rockbox, bitr
torrentscsv, xfsb, yihua, pirateiro,

test_*_guards.py, test_*_deep.py (crash guards + deep coverage variants
test_ci_infra.py, test_ci_workflows.py
test_community_plugins_compile.py
test_courses_scaffold.py, test_course_scripts_lint.py
test_docs_presence.py
test_frontend_spec_coverage.py
test_install_plugin_json_config.py
test_jackett_integration.py, test_jackett_plugin_pool.py
test_nmclub_config_selfheal.py, test_nmclub_plugin_login.py
test_no_runtime_service_skips.py
test_openapi_frozen.py
test_page_title.py, test_footer_restored.py
test_palette_catalog.py, test_palette_catalog_python_mirror.py
test_quality_compose.py, test_quality_detection.py
test_readme_landing_page.py
test_runtime_requirements_includes_new_deps.py
test_scan_script_non_interactive.py
test_scanner_configs.py
test_service_fixtures.py
test_shadow_tokens.py
test_socks_udp.py
test_start_sh_copy_plugins_framework.py
test_toolchain_config.py

```

test_tracker_validator.py
test_website_config.py
test_branding_assets.py
test_architecture_diagrams.py
test_build_releases_non_interactive.py
tests/benchmark/           (3)
tests/chaos/               (2)
tests/concurrency/         (3)
tests/contract/            (5)
tests/docs/                (3)
tests/e2e/                 (7)
tests/integration/         (19)
tests/memory/              (1)
tests/observability/       (2)
tests/performance/         (1)
tests/property/            (2)
tests/security/            (10)
tests/stress/              (2)

```

Commits (latest 10+)

```

c8254ee fix: 21 test failures, pirateiro 44/44, coverage gate 88%, dedup c
ccd0dc9 fix: env_loader flaky test + coverage baseline 88.14% + minor test
3d1819c docs: finalize Session 8 â€” Fixed.md Rev 12, 59 items, 18 bugs, 4
f924841 test: wave 8 â€” btsow/torrentscsv/xfsub/yihua (170 tests) + bt4g
6773203 test: wave 7 â€” yourbittorrent/glotorrents/pctorrent/rockbox/bitr
f191401 test: wave 6 â€” torrentfunk/torrentproject/therarbg/academicorre
1446cc5 test: wave 5 â€” audiobookbay/one337x/extratorrent (155 tests) + 3
fd543b0 docs: update CONTINUATION.md for Session 8 (4 waves, 23 plugins)
aae4069 test: wave 4 â€” kinozal/nnmclub/rutracker/megapeer/jackett + mega
e3b7acb test: wave 3 â€” rutor/tokyotoshokan/snowfl/torrentdownload/linux

```

Coverage Snapshot

Module	Coverage
api/init.py	98%
api/auth.py	91%
api/routes.py	95%
api/streaming.py	99%
main.py	94%
deduplicator.py	94%
enricher.py	100%
hooks.py	95%
jackett_autoconfig.py	99%
scheduler.py	93%
search.py	~90%
validator.py	92%
theme_injector.py	99%

Module	Coverage
env_loader.py	100%
download_proxy.py	~95%
TOTAL (unit)	88%+

Known Issues

1. **BOB-008:** RuTracker CAPTCHA “ operator-blocked (needs manual cookie paste)
 2. Go backend is a skeleton (documented in AGENTS.md)
 3. Containers may be down on session start “ bash start.sh first
 4. macOS + podman network_mode: host does NOT forward ports “ ensure-macos-tunnel.sh handles this
-

Quick-Start

```
# Pre-build gate
bash scripts/pre_build_verification.sh

# Unit tests
python3 -m pytest tests/unit/ -q --import-mode=importlib

# Coverage
python3 -m pytest tests/unit/ --cov=download-proxy/src --cov=plugins --cov=

# Lint + typecheck
ruff check . && mypy download-proxy/src/

# Frontend
cd frontend && npx vitest run

# Containers
bash start.sh && bash stop.sh
```

Architecture

Port Service	Tech
7185 qBittorrent WebUI	LinuxServer (admin/admin)
7186 Download Proxy	Python HTTP
7187 Merge Search Service	FastAPI + Angular SPA
7188 WebUI Bridge	Python (host process)
7189 boba-jackett	Go/Gin
9117 Jackett	LinuxServer (auto-configured)